

The *jing* and *ras1* pathways are functionally related during CNS midline and tracheal development

Margaret J. Sonnenfeld*, Nasrin Barazesh, Yalda Sedaghat, Cathy Fan

Department of Cellular and Molecular Medicine, Faculty of Medicine, University of Ottawa, Ottawa, Ont., Canada K1H 8M5

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Abstract

The *Drosophila jing* gene encodes a zinc finger protein required for the differentiation and survival of embryonic CNS midline and tracheal cells. We show that there is a functional relationship between *jing* and the *Egfr* pathway in the developing CNS midline and trachea. *jing* function is required for *Egfr* pathway gene expression and MAPK activity in both the CNS midline and trachea. *jing* over-expression effects phenocopy those of the *Egfr* pathway and require *Egfr* pathway function. Activation of the *Egfr* pathway in loss-of-function *jing* mutants partially rescues midline cell loss. *Egfr* pathway genes and *jing* show dominant genetic interactions in the trachea and CNS midline. Together, these results show that *jing* regulates signal transduction in developing midline and tracheal cells.

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1. Introduction

The *Drosophila jing* gene functions in multiple developmental processes and encodes a nuclear protein with zinc fingers and regions suggestive of transcriptional regulation (Sedaghat et al., 2002; Liu and Montell, 2001). In the ovary, *jing* is required for border cell migration downstream of the vertebrate homolog of the basic region/leucine zipper transcription factor CCAAT enhancer-binding protein (C/EBP) (Liu and Montell, 2001). During embryogenesis, *jing* function is required for the differentiation and survival of cells in the CNS midline, trachea and brain downstream of the master regulators *single-minded* and *trachealess* (Sedaghat et al., 2002; Sedaghat and Sonnenfeld, 2002). JING shows a nuclear localization in CNS midline glia and tracheal cells during early and late stages of differentiation suggesting that it may have a regulatory role in these cells. However, the mechanism of *jing* function and its downstream targets are unknown.

Previous studies show that tyrosine kinase signaling through the *Drosophila* epidermal growth factor receptor (EGFR) regulates midline glial differentiation and survival as well as tracheal patterning (Sonnenfeld and Jacobs, 1994; Scholz et al., 1997; Stemerink and Jacobs, 1997; Wappner et al., 1997). Tracheal activation of the *Egfr* pathway controls the formation of the dorsal trunk and visceral branches (Wappner et al., 1997). The tracheal and CNS midline phenotypes of embryos homozygous mutant for *jing* and *Egfr* pathway genes are similar suggesting that these genes may function in the same pathway (Sonnenfeld and Jacobs, 1994; Sedaghat et al., 2002). To address this and in order to understand the regulatory mechanisms that control midline glial and tracheal cell differentiation we have characterized the functional relationship between the *jing* and *Egfr* pathways during embryogenesis.

Egfr signaling is required in multiple developmental processes including *Drosophila* oogenesis, embryogenesis and adult eye development (Casici and Freeman, 1999; Wassarman and Freeman, 1997; Schweitzer and Shilo, 1997; Perrimon and Perkins, 1997). Receptor tyrosine kinases (RTKs) activate RAS1 which in turn activates protein kinases in the Ras1/mitogen activated protein kinase

* Corresponding author. Tel.: +1-613-562-5800x8150; fax: +1-613-562-5434.

E-mail address: msonnenf@uottawa.ca (M.J. Sonnenfeld).

(MAPK) signaling cascade. MAPKs phosphorylate transcription factors and result in the regulation of gene expression and cellular development. Data indicate that RTKs all act through a common pathway involving MAPK and generate a generic signal that is interpreted in responding cells according to their developmental history (Halfon et al., 2000; Simon, 2000; van der Geer et al., 1994; Gabay et al., 1997a,b).

Tissue-specific activities of the EGFR in the CNS midline and trachea are regulated by the ligand SPI, which is similar to the mammalian TGF- α (Rutledge et al., 1992). Activation of SPI requires two proteins, Rhomboid-1 (RHO-1) (a seven transmembrane serine protease) and STAR (S) (a single-pass transmembrane protein) which regulate the intracellular trafficking and proteolysis of SPI (Lee et al., 2001; Urban et al., 2001; Bier et al., 1990; Kolodkin et al., 1994; Guichard et al., 1999; Schweitzer et al., 1995a; Bang and Kintner, 2000; Wassarman et al., 2000).

Differentiation of CNS midline glia during *Drosophila* embryogenesis requires activation of EGFR (Zak et al., 1990; Raz and Shilo, 1992). EGFR is widely expressed and is activated in the midline due to expression of *rho* (Raz and Shilo, 1992; Zak et al., 1990; Gabay et al., 1997a,b). Regulation of EGFR activity in the CNS midline occurs in the presence of soluble Spi and by induction of the inhibitory protein Argos (Schweitzer et al., 1995a,b; Golembo et al., 1996a,b). Membrane-SPI is processed into secreted-SPI in CNS neurons and therefore axon contact is required for EGFR activation in midline glia (Bergmann et al., 2002). EGFR activates the conserved MAPK pathway which then activates the ETS domain transcription factor PNT and inactivates YAN (O'Neill et al., 1994; Scholz et al., 1997).

Loss or reduction of function in genes of the *Egfr* pathway leads to the premature death of the midline glia during stage 12 (Sonnenfeld and Jacobs, 1994; Scholz et al., 1997; Lanoue et al., 2000). In contrast, an increase in EGFR/ras1/raf signaling correlates with an increased survival of midline glia (Sonnenfeld and Jacobs, 1995; Scholz et al., 1997; Stermerdink and Jacobs, 1997; Lanoue and Jacobs, 1999; Dong and Jacobs, 1997). There is now strong evidence that this survival event is mediated through Ras1, which in turn inhibits apoptosis by phosphorylating head involution defective (Hid) and inactivating its proapoptotic effects (Bergmann et al., 1998, 2002; Kurada and White, 1998).

In this study, we show that the *jing* and *Egfr* pathways are functionally related in the embryonic CNS midline and trachea. The results demonstrate that *jing* is required for MAPK activity and *Egfr* pathway gene expression. *jing* over-expression can induce extra *Egfr*-expressing glia in the CNS midline and therefore resembles the over-expression phenotype of *Egfr* pathway genes (Klaes et al., 1994; Sonnenfeld and Jacobs, 1995; Stermerdink and Jacobs, 1997). Dominant genetic interactions reveal that *jing*

and *Egfr* function is critical for proper CNS midline and tracheal cell development. Together, these results show that *jing* is required for proper EGFR signaling.

2. Results

2.1. *jing* function is required for MAPK activity in the CNS midline and tracheal placodes

In homozygous *jing* mutant embryos, many cells in the CNS midline and trachea undergo apoptosis which is first detectable by TUNEL labeling during stage 12 and 10, respectively (Sedaghat et al., 2002). Since EGFR signaling is essential for the survival of midline glia we thought it may be possible that EGFR-mediated RAS1 signaling is down-regulated in *jing* mutant embryos (Scholz et al., 1997; Sonnenfeld and Jacobs, 1994; Lanoue et al., 2000). The activity of the Ras1 pathway was investigated in the absence of *jing* function by monitoring the presence of the activated, diphosphorylated form of MAPK (ERK) in CNS midline and tracheal cells in homozygous *jing* mutant embryos. These embryos were stained with a monoclonal antibody that recognizes diphospho-ERK representing the total activities of RTK signaling in *Drosophila* embryos (Gabay et al., 1997a,b).

Wild-type and homozygous *jing* mutant embryos were stained with anti-dp-ERK and anti-TRH to identify tracheal cells or anti-SIM to identify midline cells. At stage 10, MAPK was strongly activated in SIM-positive midline cells in wild-type embryos but not in homozygous *jing* mutant embryos (Fig. 1A–B'). It is possible that the MAPK reductions are not due to cell death since there were no defects in midline cell numbers and nuclear size in *jing* mutants with reduced MAPK (Fig. 1B,B'). MAPK activity was also reduced in the ventral ectoderm in regions where midline MAPK activity was weak, consistent with its dependence on active SPI from the midline (Golembo et al., 1996b; Schweitzer et al., 1995a; Gabay et al., 1997a).

Activation of EGFR by SPI in the tracheal placodes is required for formation of the dorsal trunk and visceral branch of the tracheal network (Wappner et al., 1997; Gabay et al., 1997a). During stage 10, MAPK activity in the tracheal placodes is dependent upon proper *Egfr* pathway function (Gabay et al., 1997b). EGFR-dependent MAPK activity in the tracheal placodes was reduced in homozygous stage 10 *jing* mutant embryos (Fig. 1B, arrowhead). During stage 11, MAPK activity is dependent on the FGF receptor, *breathless* (*btl*) in wild-type embryos (Gabay et al., 1997b; Glazer and Shilo, 1991; Klämbt et al., 1992; Reichman-Fried and Shilo, 1995) but MAPK was not activated in homozygous *jing* mutant embryos (Fig. 1C–D'). Therefore, *jing* function is required during both *Egfr*- and *btl*-dependent stages of MAPK activity in the developing trachea.

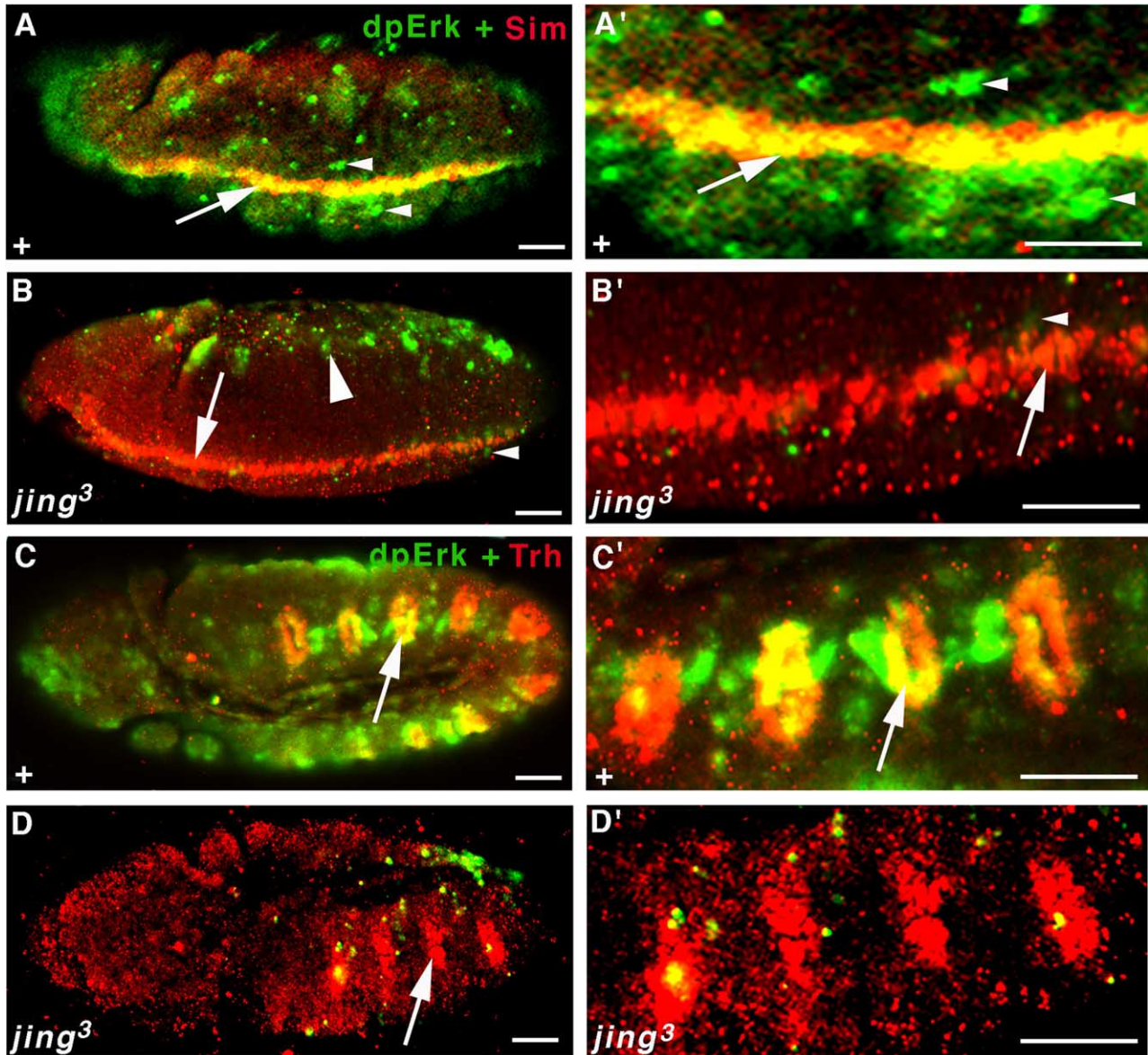


Fig. 1. *jing* is required to maintain MAPK activity in CNS midline and tracheal cells. Whole-mount embryos were double labeled with anti-dpErk and anti-SIM (A–B') or anti-trachealess (TRH) (C–D'). (A) Midline cells show MAPK activity at stage 10 (arrow). The adjacent ventral ectoderm also shows MAPK activity (arrowheads). (A') Close-up view of the midline cells in (A) showing that the majority of cells express MAPK (arrow). (B,B') A stage 10 homozygous *jing*³ mutant embryo with reduced MAPK activity in the midline (arrow) and ventral ectoderm (small arrowhead). In segments containing more MAPK in the midline (B', arrow) there is more MAPK in the ventral ectoderm (arrowhead) consistent with the inductive capabilities of midline cells. The large arrowhead in (B) denotes reduced EGFR-dependent MAPK activity in the tracheal pits. (C) A wild-type stage 11 embryo showing MAPK activity in the tracheal pits (arrow). (C') Close-up view of embryo in (C) showing that many TRH⁺ cells have MAPK activity. (D) An homozygous stage 11 *jing*³ mutant embryo showing a severe phenotype. Most TRH⁺ cells do not have *btl*-dependent MAPK activity in the tracheal pits (arrow). (D') Close-up view of the embryo in (D). All images are merged. Scale bars: 30 μ m.

2.2. *jing* is required for *Egfr* pathway gene expression

To determine whether *jing* is required for *Egfr* pathway transcription, we analyzed the mRNA levels of *Egfr* pathway genes in *jing* homozygous mutant embryos and after over-expression in the CNS midline and trachea. As shown by microarray analysis, the levels of *ras1* mRNA were reduced by 50% in *jing* homozygous mutant embryos compared to the levels of control *w*¹¹⁸

and midline glial-expressed genes *wrapper* and *asteroid* (*ast*) ($P < 0.05$) (Fig. 2A) (Kotarski et al., 1998; Noordermeer et al., 1998). Other genes expressed in the CNS midline (such as *center divider*, *cdi*) or in the trachea (such as *thick veins*, *ttv*), also showed reduced mRNA levels in comparison to control levels (Affolter et al., 1994; Matthews and Crews, 1999).

When *jing* was over-expressed in the CNS midline and trachea, using the *Gal4/UAS* targeted gene expression

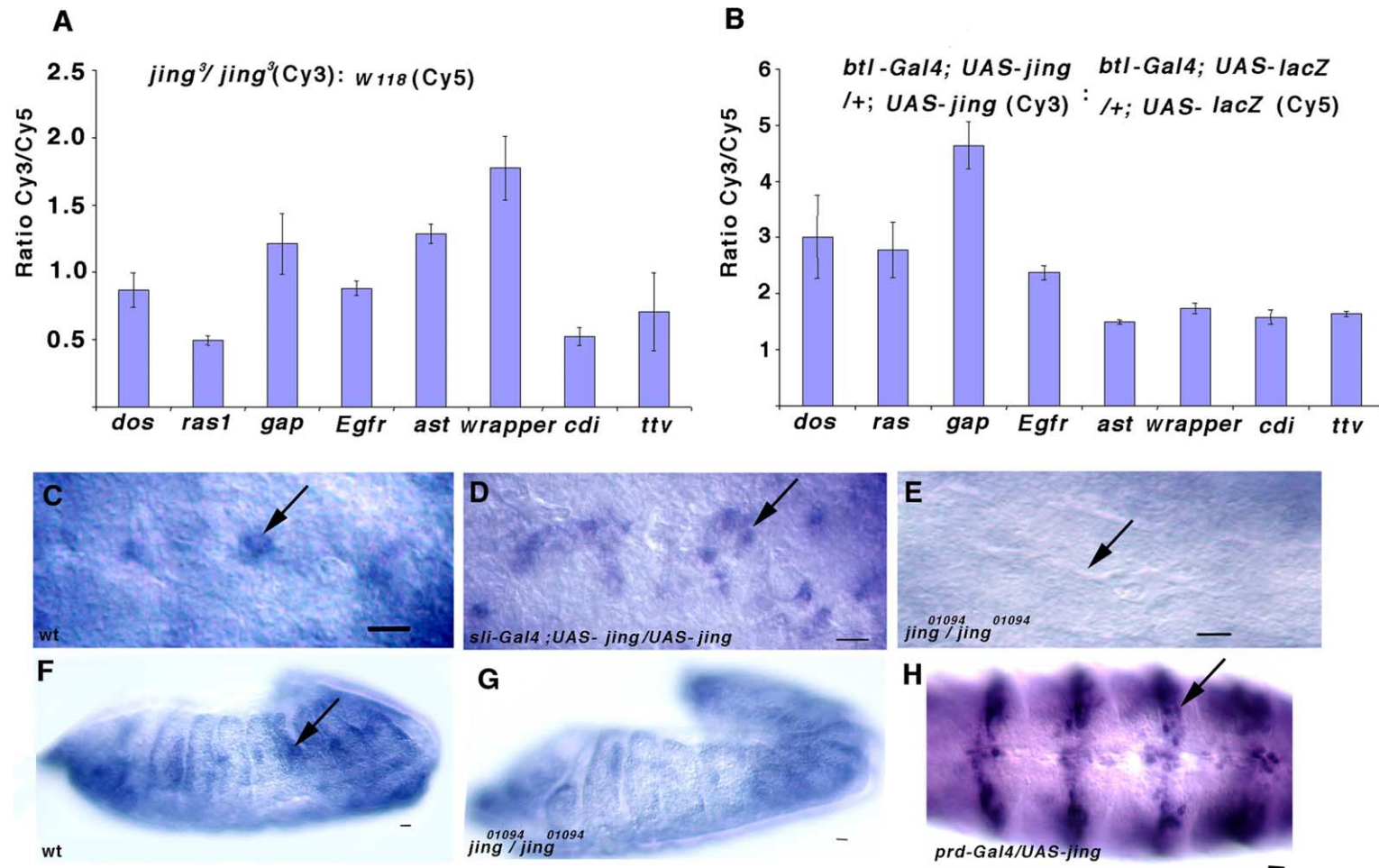


Fig. 2. Effects of *jing* on *Egfr* pathway gene expression. (A,B) Microarray analysis of total RNA extracted from embryos homozygous mutant for *jing*³ (A) and over-expressing *jing* in CNS midline glia and trachea (B). Ratios of experimental (Cy3) to control (Cy5) mRNA fluorescent intensities are shown. (A) In homozygous *jing* mutants, the levels of *ras* and *cdi* mRNA are most significantly reduced (50% reduction, $P < 0.05$) from control (*w*¹¹⁸) levels. (B) *jing* was over-expressed in CNS midline glia and in the trachea using the *btl-Gal4* driver. mRNA levels of all indicated *Egfr* pathway genes increased by at least two-fold over control levels and over those of other midline or tracheal-expressed genes ($P < 0.05$). (C–G) Whole-mount in situ hybridization using an *Egfr* riboprobe. Anterior is to the left with ventral views (C–E,H) and sagittal views of the trachea (F,G). (C) In wild-type stage 12/3 embryos, *Egfr* is expressed in 1–2 midline glia (arrow). (D) There is an increase in the number of *Egfr*-expressing glia (arrow) after *jing* over-expression in midline glia, as driven by *sli-Gal4*. (E) There is an absence of CNS midline *Egfr* expression (arrow) in stage 12/3 homozygous *jing*⁰¹⁰⁹⁴ mutant embryos. (F) *Egfr* expression in the trachea (arrow) of control stage 12 embryos. (G) *Egfr* expression is reduced in the trachea of *jing*⁰¹⁰⁹⁴ homozygous mutant stage 12 embryos. (H) Strong expression of *jing* mRNA in *prd*-expressing stripes in a stage 13 embryo carrying P[*prd-Gal4*] and P[*UAS-jing*]. Scale bars: 10 μ m.

system and the *btl-Gal4* driver (Brand and Perrimon, 1993; Shiga et al., 1996), small but statistically significant two-fold increases in *Egfr* pathway mRNA levels were observed except for *gap* which increased almost four-fold ($P < 0.05$) (Fig. 2B). Other CNS midline- or tracheal-expressed genes, such as *ttv*, *ast*, *wrapper* and *cdi*, did not show significant changes in mRNA levels in comparison to controls after *jing* over-expression (Fig. 2B).

To further address whether *jing* activates *Egfr* expression, embryos homozygous mutant for *jing* and those over-expressing *jing* were subjected to in situ hybridization using an *Egfr* riboprobe. As shown in Fig. 2D, extra *Egfr*-expressing glia were observed when *jing* was over-expressed in the CNS midline using a *Gal4* fusion to the promoter of the *slit* (*slit*) gene (*slit-Gal4*). *slit-Gal4* drives expression specifically in midline glia from stage 11 until the end of embryogenesis (Scholz et al., 1997; Wharton et al., 1994). Similar results were obtained after driving *jing* expression in all CNS midline cells using a *sim-Gal4* driver (data not shown). Conversely, there were reductions in midline *Egfr* expression in homozygous *jing*⁰¹⁰⁹⁴ and *jing*³ (data not shown) embryos (Fig. 2E). Tracheal *Egfr* expression was also reduced in *jing* homozygous mutant embryos (Fig. 2G) compared to controls (Fig. 2F). *jing* function may thus be considered essential for *Egfr* pathway gene expression in both the CNS midline and trachea.

Ectopic expression studies were used to determine whether *jing* can ectopically activate the *Egfr*. *jing* was expressed in the ectodermal stripes of the *paired* (*prd*) gene using a *prd-Gal4* driver and one copy of *UAS-jing*, as this was sufficient to obtain strong *jing* activation (Fig. 2H). In situ hybridization using an *Egfr* riboprobe failed to detect any mRNA expression in *prd*-expressing cells suggesting that *jing* cannot ectopically activate *Egfr* transcription (data not shown).

2.3. Over-expression of *jing* in the CNS midline phenocopies gain-of-function phenotypes in EGFR signaling

Over-expression of *Egfr* pathway genes in the CNS midline leads to supernumerary glia and in some cases, to lack of commissural connections (Klaes et al., 1994; Sonnenfeld and Jacobs, 1995; Stemerink and Jacobs, 1997; Scholz et al., 1997). We, therefore, reasoned that if *jing* regulates the *Egfr* pathway, then over-expression of *jing* should result in a similar phenotype. To address this, *jing* was over-expressed in the midline glia by crossing flies carrying two copies of a *UAS-jing* transgene to flies carrying the *slit-Gal4* driver. The progeny from this cross were stained with anti-Sim to determine the effect of *jing* over-expression on both midline neurons and glia. In addition, a reporter for *slit* expression (*slit-lacZ*) was used to determine the number of glia per VNC segment after *slit-Gal4*-induced *jing* over-expression. *slit-lacZ* is expressed specifically in

midline glia from stage 11 until the end of embryogenesis (Ma et al., 2000).

Increases in midline cell numbers were observed after *jing* over-expression (Fig. 3B,E,H). Quantitative analysis revealed an average of 16 (± 1) SIM-positive cells after *jing* over-expression ($n = 30$ segments) compared to 11 (± 2) in each VNC segment of wild-type stage 13 embryos ($n = 45$ segments) representing a statistically significant difference ($P < 0.05$) (Fig. 3B). Similar results were found using an independent *UAS-jing* strain (data not shown). The extra cells expressed *slit-lacZ* confirming their glial identities while *slit-Gal4* heterozygotes did not induce extra *slit-lacZ* expression (Fig. 3E). There was an average of 12 glia that were scattered laterally from the midline compared to eight in wild-type embryos ($P < 0.05$) (Figs. 3C–E,4). Interestingly, the supernumerary glia were still located dorsally in the VNC (Fig. 3F), similar to those found after ectopic expression of *pointed*^{P1} (*pnt*^{P1}) in the midline (Klaes et al., 1994). There was no cell death present in the glial population (Fig. 3F) as compared to the small apoptotic glia found surrounding the nerve cord in wild-type embryos (Fig. 3D, arrowheads) (Sonnenfeld and Jacobs, 1995). These results suggest that *jing* over-expression suppresses normal midline glial cell death and correlates with the increase in *Egfr*-expressing glia (see Fig. 2D).

CNS axon formation is a sensitive indicator of midline cell function (Klämbt et al., 1991; Klämbt and Goodman, 1991). To determine whether the extra glia affected CNS axon formation, embryos over-expressing *jing* were stained with the CNS axon specific monoclonal antibody BP102. *jing* over-expression in midline glia was associated with a lack of commissural fascicle formation similar to its over-expression in all CNS midline cells (Fig. 3G,H) (Sedaghat et al., 2002). There was an association between the laterally dispersed supernumerary glia and the lack of commissure formation as shown in Fig. 3H (arrow). The supernumerary glia appeared to migrate towards the longitudinal axons as they have previously been observed to do in *commissureless* mutants and after ectopic expression of *pnt*^{P1} in the midline (Klaes et al., 1994; Sonnenfeld and Jacobs, 1995). In summary, *jing* is able to induce supernumerary glia in a similar fashion as the *pnt*^{P1} transcription factor.

2.4. *jing* gain-of-function is suppressed by *Egfr* loss-of-function

The genetic data indicates that *jing* may have an upstream function in the *Egfr/ras1* pathway. To address this issue, the effect of a reduction in EGFR signaling on the *jing* gain-of-function phenotype was examined in the midline glia. The *sim-Gal4* and *slit-Gal4* drivers were used to over-express *jing* specifically in the CNS midline in heterozygous and homozygous *spi* and *S* mutant backgrounds. The number of *slit-lacZ*-expressing midline glia in each nerve cord segment was quantified during stage 13 and compared to that in wild-type embryos over-expressing *jing*

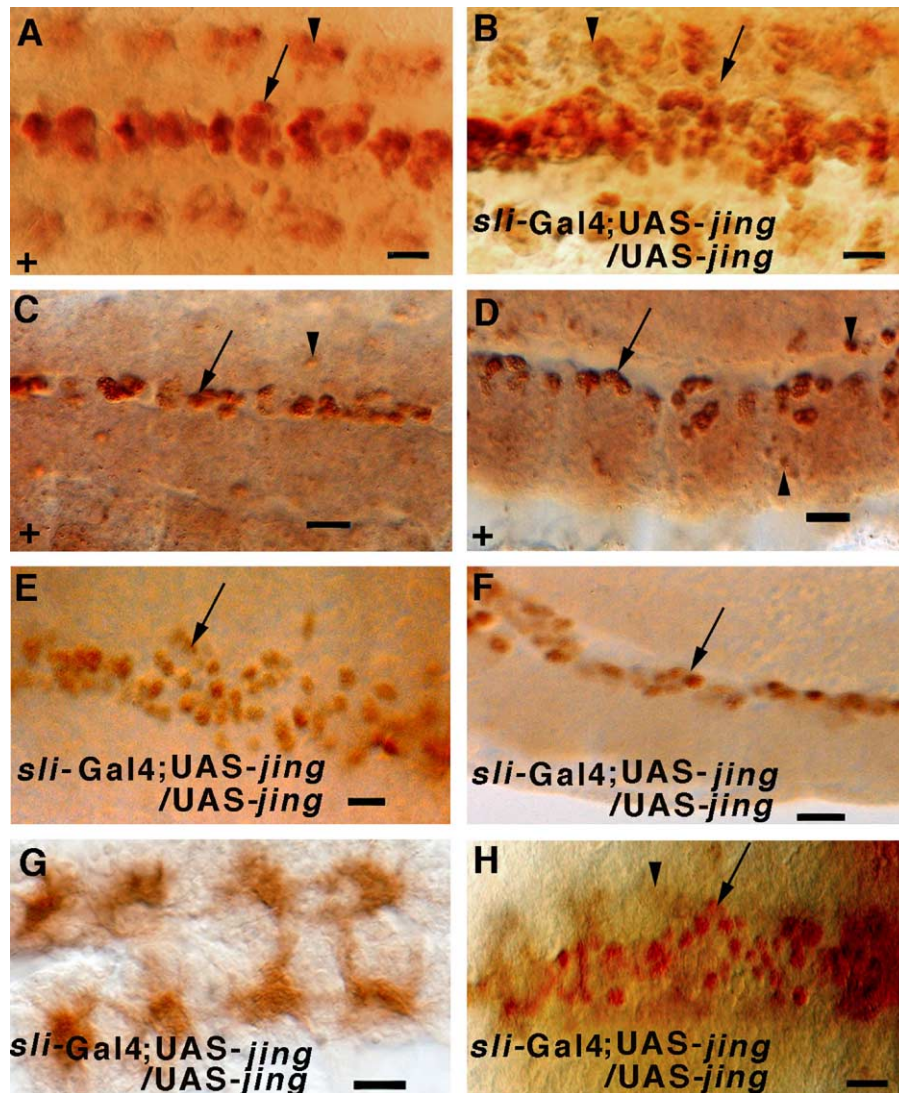


Fig. 3. *jing* over-expression in midline glia alters cell numbers and commissure formation. Two copies of *UAS-jing* were placed under control of *sli-Gal4* and embryos were stained with anti-SIM to identify midline neurons and glia (A,B), with anti- β -Galactosidase to assess *sli-lacZ* expression (C–F,H) or with BP102 to visualize CNS axons (G,H). Ventral views (A–C,E,G,H) and sagittal views (D,F) of stage 13 embryos are shown with anterior to the left. (A) In wild-type embryos, SIM localizes to glia and neurons along the midline (arrow) and laterally to muscle precursors (arrowhead). (B) In an embryo over-expressing *jing* in midline glia, SIM localizes to additional ectopically distributed midline cells (arrow). The pattern of SIM staining in the muscle precursors is unaltered from wild-type (arrowhead). (C,D) In wild-type embryos, *sli-lacZ* expression is detected in approximately 6–7 glia along the midline in each segment (C, arrow) and in dorsal positions in the nerve cord (D, arrow). Additional small *sli-lacZ*-expressing cells outside the ventral and dorsal regions of the nerve cord are apoptotic glia (arrowheads). (E,F) After *sli-Gal4*-directed expression of *jing*, *sli-lacZ* is expressed in an average of twelve glia (arrow) that are dispersed laterally from the midline. The extra glia maintain a dorsal position in the nerve cord (F, arrow). Note the absence of *sli-lacZ*-expressing apoptotic glia surrounding the nerve cord. (G) Commissural axons do not properly approach the midline after *sli-Gal4*-directed expression of *jing*. The axons instead appear stalled along the longitudinal tracts. (H) A stage 12/5 embryo over-expressing *jing* in the CNS midline glia and carrying *sli-lacZ*. Arrowhead denotes CNS axons. The supernumerary glia migrate toward longitudinal axons (arrow). *n* is greater than 70 segments. Scale bars: 10 μ m.

(Fig. 4). Expression of two copies of the *UAS-jing* transgene in the midline glia of wild-type or heterozygous *spi* and *S* embryos resulted in an average of 12 midline glia instead of the normal 8 during stage 13 (Fig. 4; $n > 50$ segments, $P < 0.05$). In contrast, *UAS-jing* transgene expression was unable to induce 12 midline glia in homozygous *spi* and *S* mutant backgrounds (Fig. 4; $n > 52$ segments, $P < 0.05$). In these embryos, there was an average of 1.5 midline glia in each nerve cord segment after *jing* over-expression which is similar to the number of midline glia present in homozygous

spi and *S* mutant embryos during stage 13 (Fig. 4) (Sonnenfeld and Jacobs, 1994).

To test the independent activity of *jing*, we chose to examine the effects of ectopic *jing* expression in the *Drosophila* eye which is a system that is functional for the *Egfr* pathway but not for *jing* or upstream regulators including *single-minded* (*sim*) or *tracheless* (*trh*) (Simon et al., 1991; Dickson et al., 1996). Analysis of *jing*⁰¹⁰⁹⁴ enhancer trap *lacZ* expression and of endogenous mRNA expression by in situ hybridization showed that *jing* is not

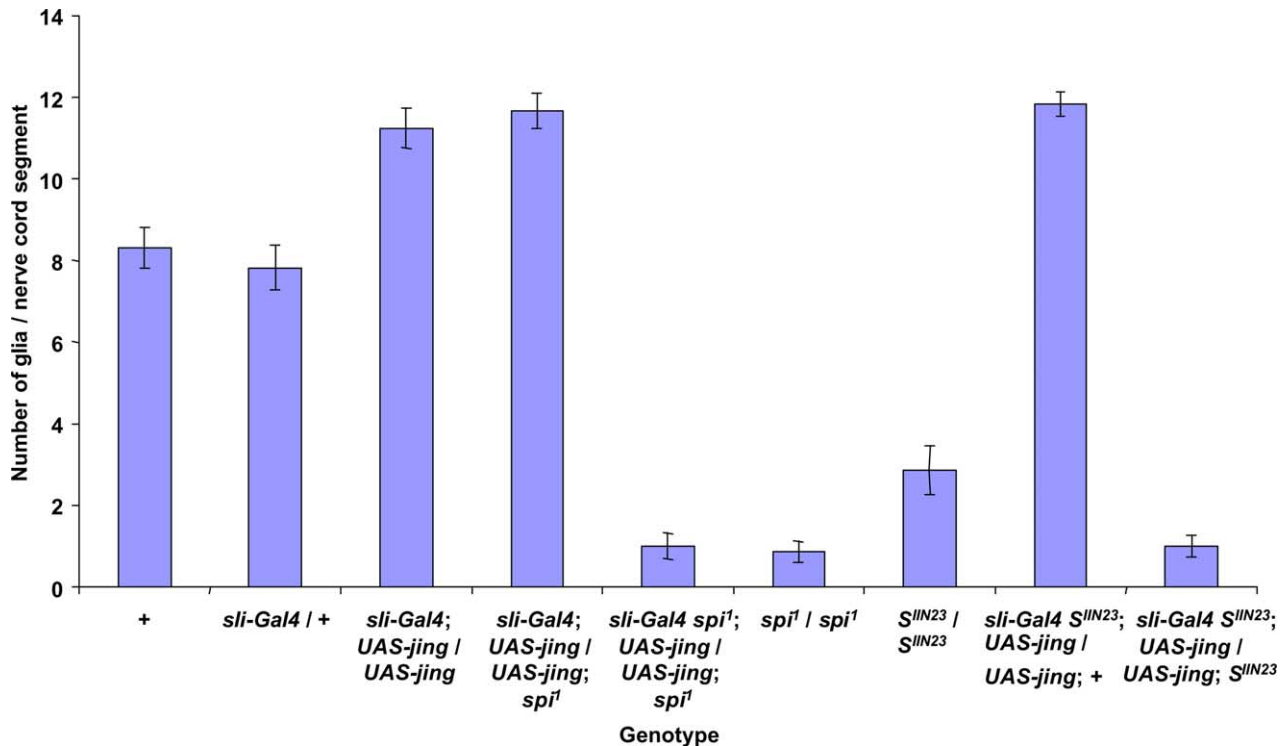


Fig. 4. Induction of extra midline glia by *jing* over-expression is suppressed by reducing EGFR signaling. The *UAS-jing* transgene was expressed specifically in midline glia using the *slit* (*sli*) promoter (*sli-Gal4*). Midline glia were detected by *sli-lacZ* expression. Expression of two copies of *UAS-jing* induces 12 midline glia compared to the normal 8 glia present in wild-type stage 13 embryos. Induction of the extra glia is inhibited by the presence of homozygous but not heterozygous *spi* and *S* mutations. The induction of extra glia by *jing* over-expression is therefore dependent on *Egfr* pathway function. In all cases, $P < 0.05$ by unpaired *t* tests in comparison to *w*¹¹⁸ and *sli-Gal4*/+ where *n* is greater than 100 segments.

expressed in third instar larval eye imaginal discs (data not shown). Expression of wild-type *jing* in the eye, under regulation of the *glass* promoter (P[GMR-*Gal4*]), was associated with a rough appearance (Fig. 5C) compared to P[GMR-*Gal4*] heterozygotes or wild-type (Fig. 5A; data not shown). The rough eye consisted of highly disorganized ommatidia and mechanosensory bristles in 45% of flies (Fig. 5C,D,H) and the number of ommatidia was reduced by 50% from that in wild-type and P[GMR-*Gal4*] heterozygous eyes (Fig. 5G). Therefore, the gain-of-function phenotypes of *jing* and *Egfr* both result in a significant reduction in ommatidia (Fortini et al., 1992; Baker and Rubin, 1989). Consistent with similar pathways, the rough eye phenotype of *Egfr* gain-of-function was not enhanced by that of *jing*. Out of 1000 flies scored, carrying P[GMR-*Gal4*] and both *UAS-jing* and *UAS-ellipse*, 100% showed the same eye phenotype as flies carrying only P[GMR-*Gal4*] and *UAS-ellipse*.

The *jing* ectopic expression phenotype was dominantly suppressed by a 50% reduction in the levels of *spi*(*spi*¹) and Df(2L)TW50 or *Egfr* deficiency (Df(2R)*Egfr*5) (Fig. 5E–H). After *spi* reduction, ommatidia were more organized and more abundant, although the position of the photoreceptors was not like that in controls (Fig. 5F,G). This interaction was not influenced by activation of the *glass* promoter in the heterozygous *spi*

background (P[GMR-*Gal4*]/*spi*¹) (Fig. 5G). These results suggest that there is a dosage-sensitive interaction between the *Egfr* pathway and *jing* function in the eye, where increased *jing* activity can be suppressed by a reduction in downstream components such as *spi* and *Egfr*. Given that *sim* and *trh* are not expressed in third instar larval eye discs (data not shown), these experiments suggest that *jing* can have an effect on the *Egfr* pathway in the absence of *sim* or *trh* and supports our model that *jing* works as an independent regulator in bHLH-PAS pathways (Sedaghat et al., 2002).

2.5. Modulating the *Egfr* pathway has strong effects on *jing*-induced midline cell death

If the Ras1/MAPK pathway is involved in midline cell survival downstream of *jing* then it may be possible to rescue midline cell reductions in *jing* mutants by providing a supply of activated RAS1 to the CNS midline. To address this, the number of embryos with reductions in midline cells was determined in control and homozygous or transheterozygous *jing* mutant embryos after expression of the *UAS-ras*^{V12} and *UAS-ellipse* transgenes. The *Egfr* pathway was activated specifically in the CNS midline using *sim-GAL4* to drive expression and *UAS-lacZ* to visualize the midline cells.

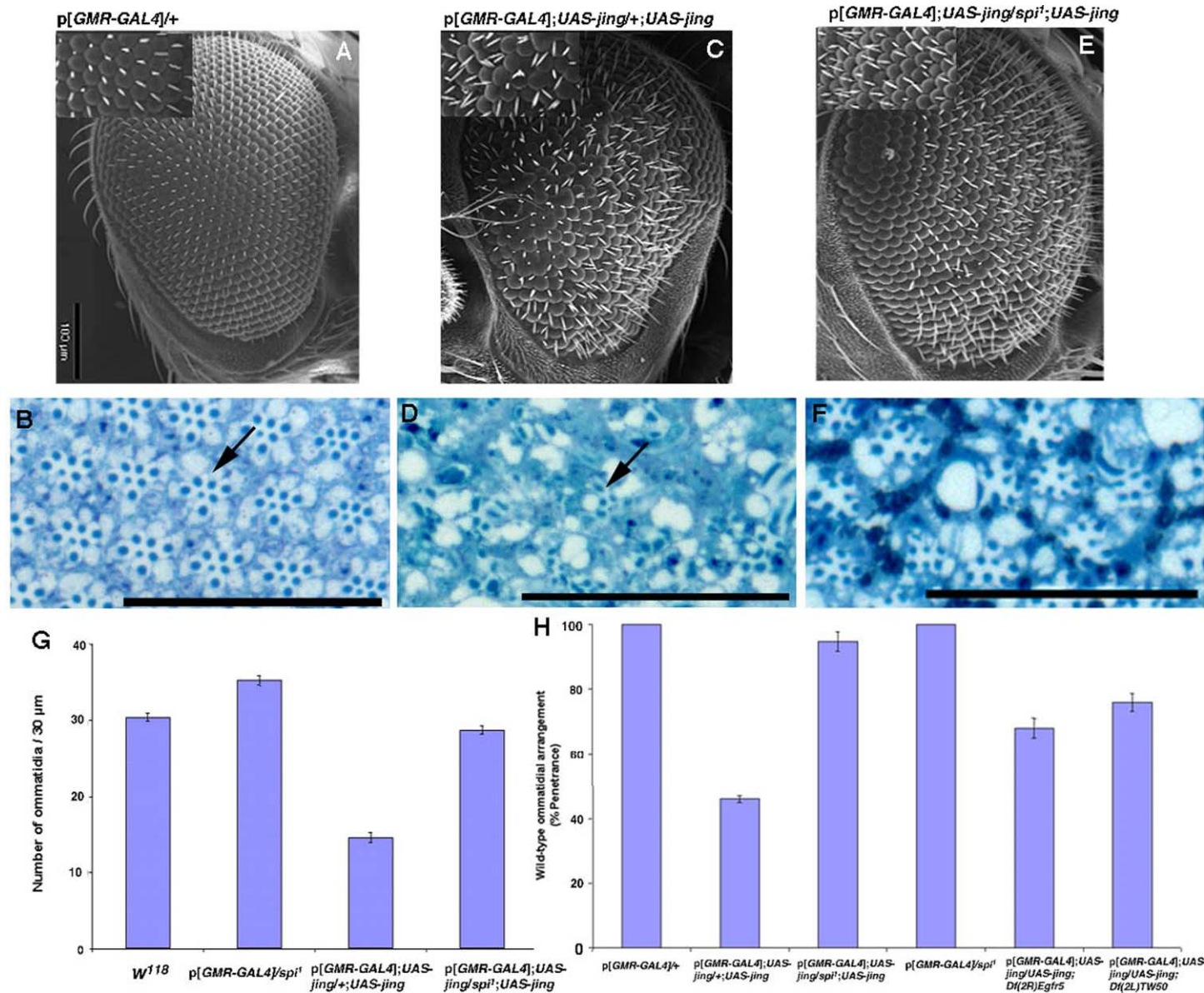


Fig. 5. EGFR pathway components lie genetically downstream of *jing* in the adult eye. (A–F) Scanning and light microscopic analysis of eyes from adults expressing the *UAS-jing* transgene under control of the *glass* promoter (P[GMR-*Gal4*]). (A,B) Flies heterozygous for P[GMR-*Gal4*] have a wild-type arrangement of ommatidia (B, arrow) and mechanosensory bristles (A, inset). (C,D) Expression of wild-type *jing* under control of the *glass* promoter causes a rough appearance. There is a disruption in the structure and arrangement of the ommatidia (D, arrow) and mechanosensory bristles (C, inset). (E,F) The *jing* over-expression phenotype is dominantly suppressed by heterozygosity for the *spi* allele. Note the less severe disruption in ommatidial structure and arrangement (F). (G) Quantification of ommatidia. (H) Quantification of *spi*, *Egfr* and *jing* interactions. Forty-five percent of flies over-expressing *jing* have a rough eye phenotype. Heterozygosity for *spi* (*spi*¹) or *spi* deficiency (Df(2L)TW50) and *Egfr* (Df(2R)*Egfr*⁵) deficiency suppresses the *UAS-jing*-mediated rough eye phenotype. Scale bars in (B), (D) and (F): 0.05 mm.

Loss-of-function alleles of *jing* are associated with a reduced number of midline cells by stage 9 compared to control embryos (Sedaghat et al., 2002). By stage 14, an absence of *sim-GAL4*-expressing midline glia was apparent in homozygous *jing* mutants compared to control embryos (Fig. 6B–E). 37.8% of homozygous *jing* mutant embryos displayed the midline cell-loss phenotype (Fig. 6A).

Expression of one copy of the *UAS-ras^{V12}* and *UAS-ellipse* transgenes suppressed the *jing* glial-loss phenotype so that only 11.3 and 20% of mutant embryos displayed reductions in midline glia, respectively (Fig. 6A). Midline-targeted expression of the *UAS-ras^{V12}* and *UAS-ellipse* transgenes was not associated with glial reductions in the absence of the *jing* mutation (Fig. 6A). Therefore, these results support a role for *jing* upstream of EGFR signaling in CNS midline glia.

2.6. The *jing* and *Egfr* pathways display dominant genetic interactions in the CNS midline and trachea

Gene dosage experiments were used to determine the effects of simultaneously altering the levels of *jing* and genes of the *Egfr* pathway. Mutations in *spi* and its regulator *Star*, have been previously characterized for their midline and tracheal phenotypes (Sonnenfeld and Jacobs, 1994; Wappner et al., 1997). To determine whether *jing* and *Egfr* function is inter-dependent, the development of the CNS midline and trachea was analyzed in double heterozygotes of *jing* and *S* or *spi*. The basis for this experiment is that if the *Egfr* and *jing* pathways are inter-dependent then simultaneous reduction of only one copy of each gene should alter CNS midline and tracheal function. Multiple *jing* alleles balanced with *wg-lacZ* *Cyo* were crossed to *S^{ΔIN23}/wg-lacZ*

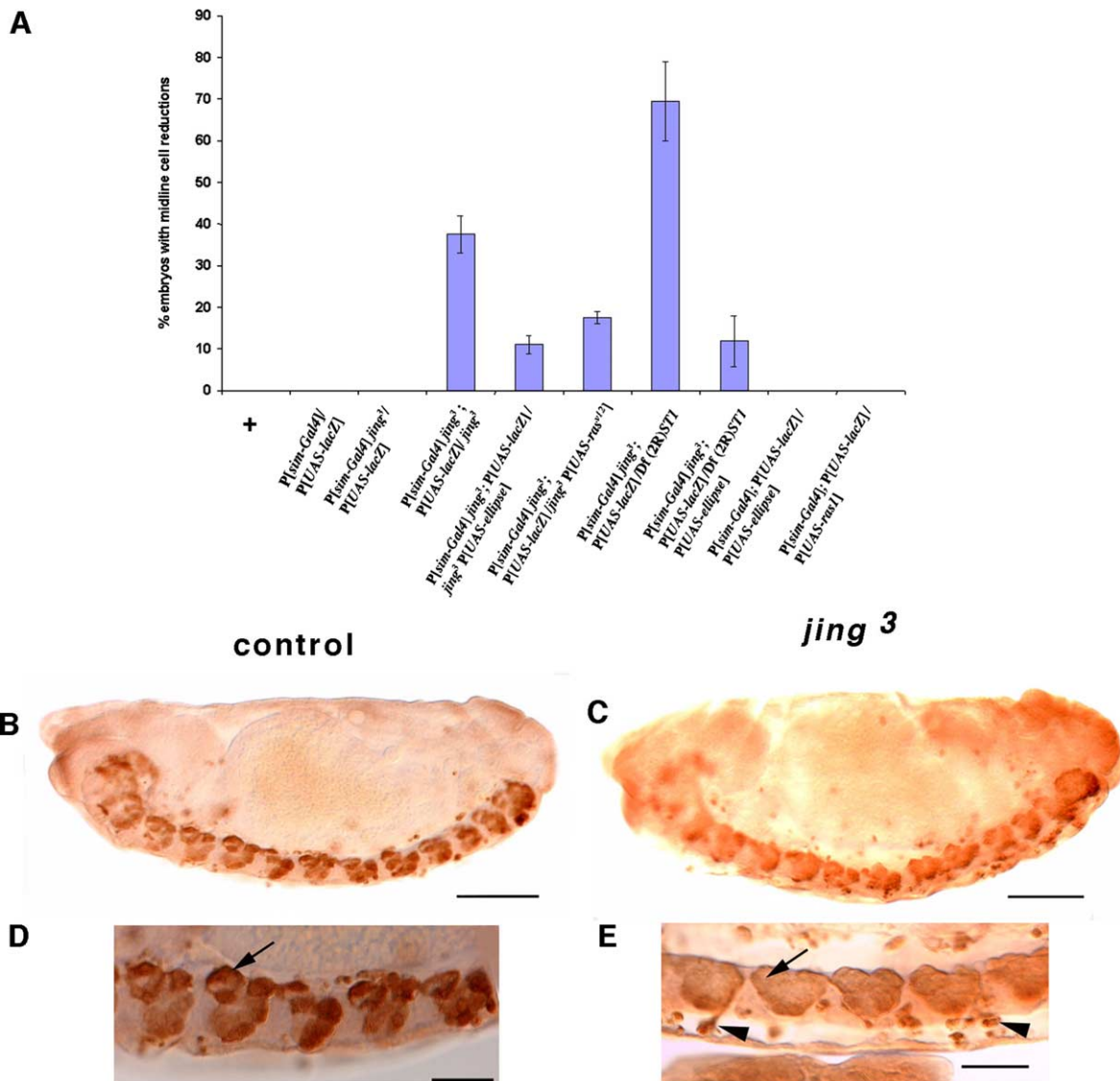


Fig. 6. (B–E) Wild-type (B,D) and *jing*³ homozygous (C,E) embryos carrying *sim-Gal4/UAS-lacZ*. Activation of the *Egfr* pathway allows midline cell survival in homozygous *jing* mutant embryos. Activated *ras* (*UAS-ras^{V12}*) and dominant *Egfr* (*UAS-ellipse*) were expressed specifically in the midline cells of homozygous *jing* mutant embryos using the *sim-Gal4* driver. Approximately 37.8% of *jing* mutant embryos show reductions in the number of midline glia. However, after expression of activated *ras* and *Egfr* (*ellipse*), only 11.3 and 20% of homozygous *jing* mutant embryos display midline glial loss. Expression of *UAS-ras^{V12}* or *UAS-ellipse* in a wild-type background is not associated with midline cell reductions.

Cyo or *spi*¹/*wg-lacZ* *Cyo* flies and their progeny were double stained with anti-SIM or anti-TRH and anti- β -Gal.

The number of CNS midline cells was reduced from wild-type in embryos homozygous and double heterozygous for *jing*, *spi* or *S* and stained with anti-SIM (Fig. 7B–D,F). Since some of the SIM-positive nuclei appeared to be fragmenting (Fig. 7D) their fate was determined by TUNEL labeling to identify apoptotic cells. In wild-type embryos, cell death is uncommon in the CNS midline during stage 12 with an average of 6($\pm$ 2) SIM-positive apoptotic nuclei per embryo (n =30). In contrast, in homozygous *jing* stage 12 mutant embryos, there was an average of 35($\pm$ 3) SIM-positive apoptotic nuclei per embryo, therefore, displaying a significant increase over that in wild-type embryos (n =35;

$P<0.05$) (Sedaghat et al., 2002). In embryos double heterozygous for mutations in *jing* and *S* or *spi* there was an average of 25($\pm$ 2) and 30($\pm$ 3) SIM-positive apoptotic nuclei per embryo during stage 12, respectively (Fig. 7E) (n =28, n =32; $P<0.05$). This is consistent with the time period for the requirement of *Egfr* function in CNS midline glia (Sonnenfeld and Jacobs, 1994; Scholz et al., 1997). Embryos heterozygous for either *jing*, *spi* or *S* mutations did not alter the normal events of midline cell apoptosis (data not shown). In summary, these results suggest that proper dosage of both *jing* and *spi* group gene function is required for midline cell survival.

In the stage 15 trachea, the number of TRH+ nuclei was reduced within each segment of embryos homozygous

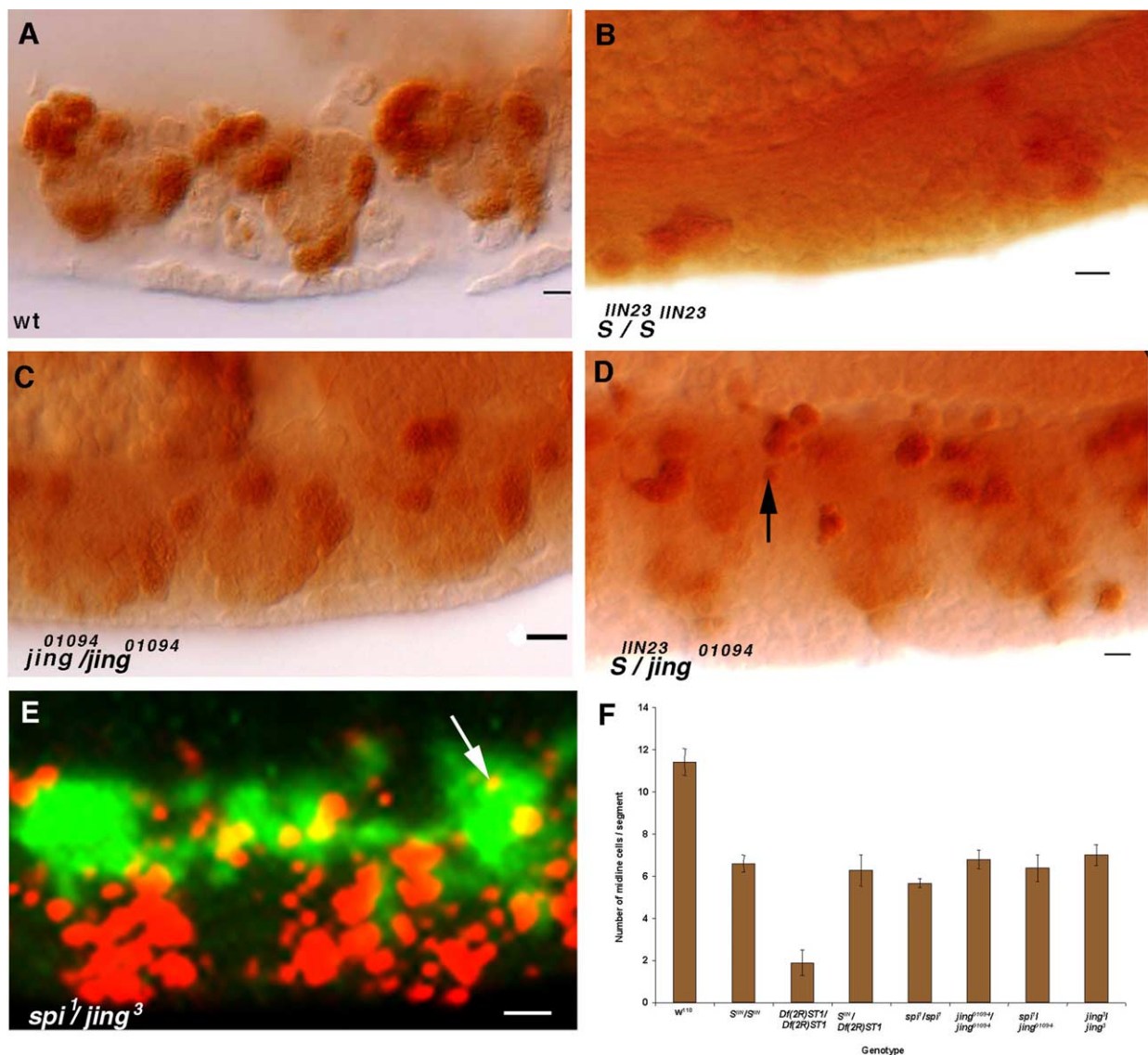


Fig. 7. The *jing* and *Egfr* pathways display dominant genetic interactions in the CNS midline. Midline cells were labeled with anti-SIM. Sagittal views of stage 14 VNCs are shown. (A) A wild-type embryo. (B) An embryo homozygous for the *S*^{11N23} mutation showing an absence of glia. (C) An embryo homozygous for *jing*⁰¹⁰⁹⁴ showing reduced SIM+ cells. (D) An embryo double heterozygous for *S*^{11N23} and *jing*⁰¹⁰⁹⁴ with SIM+ cell reductions. Fragmentation of nuclei is apparent (arrow). (E) Embryos double heterozygous for *spi* and *jing* have TUNEL+ SIM+ cells (arrow). (F) Quantification of midline cells. Scale bars: 10 μ m.

and double heterozygous for *jing* and *spi* or *S*, (Fig. 8A). In addition, breaks were observed in the dorsal trunk of double heterozygotes (Fig. 8E,F,H), consistent with the homozygous phenotypes of *jing* and *Egfr* pathway mutants (Fig. 8C,D) (Sedaghat et al., 2002). In *jing* homozygous and transheterozygous mutants, there also were truncated transverse connectives suggesting that *jing* is not only involved in *Egfr* signaling in the trachea (Fig. 8C, arrowheads; data not shown) (Wappner et al., 1997).

In embryos double heterozygous for *jing* and *S* or *spi* mutations, apoptotic tracheal nuclei were detected in stage 15 embryos (Fig. 8H,J; data not shown). The overall embryonic pattern of apoptosis was not significantly altered from wild-type suggesting that the tracheal defects did not result from gross perturbations to development (compare Fig. 8G,H). Embryos heterozygous for *jing*, *spi* or *S* did not show any alterations from wild-type in tracheal cell numbers (data not shown).

3. Discussion

3.1. *jing* and *Egfr/Ras1* function is essential for midline cell differentiation and survival

The *jing* gene was identified in two independent genetic screens for regulators of CNS midline development and border cell migration, two processes which are regulated by the EGFR (Scholz et al., 1997; Sedaghat et al., 2002; Liu and Montell, 2001; Duchek and Rorth, 2001). RTK signaling pathways have been implicated in multiple cell biological processes including proliferation, migration, differentiation and survival (Baonza et al., 2002; Duchek and Rorth, 2001; Wassarman and Therrien, 1997; Casci and Freeman, 1999; Kurada and White, 1998). How one MAPK pathway controls such different outcomes is a major area of research. Studies of *Egfr* function in the *Drosophila* adult eye suggest that signaling levels dictate the multiple cellular responses to the EGFR, such that differentiation requires the highest levels of signaling while mitosis and cell survival require less (Yang and Baker, 2003). Therefore, it is important to understand the mechanisms that control the expression of positive and negative regulators of this family of signaling molecules.

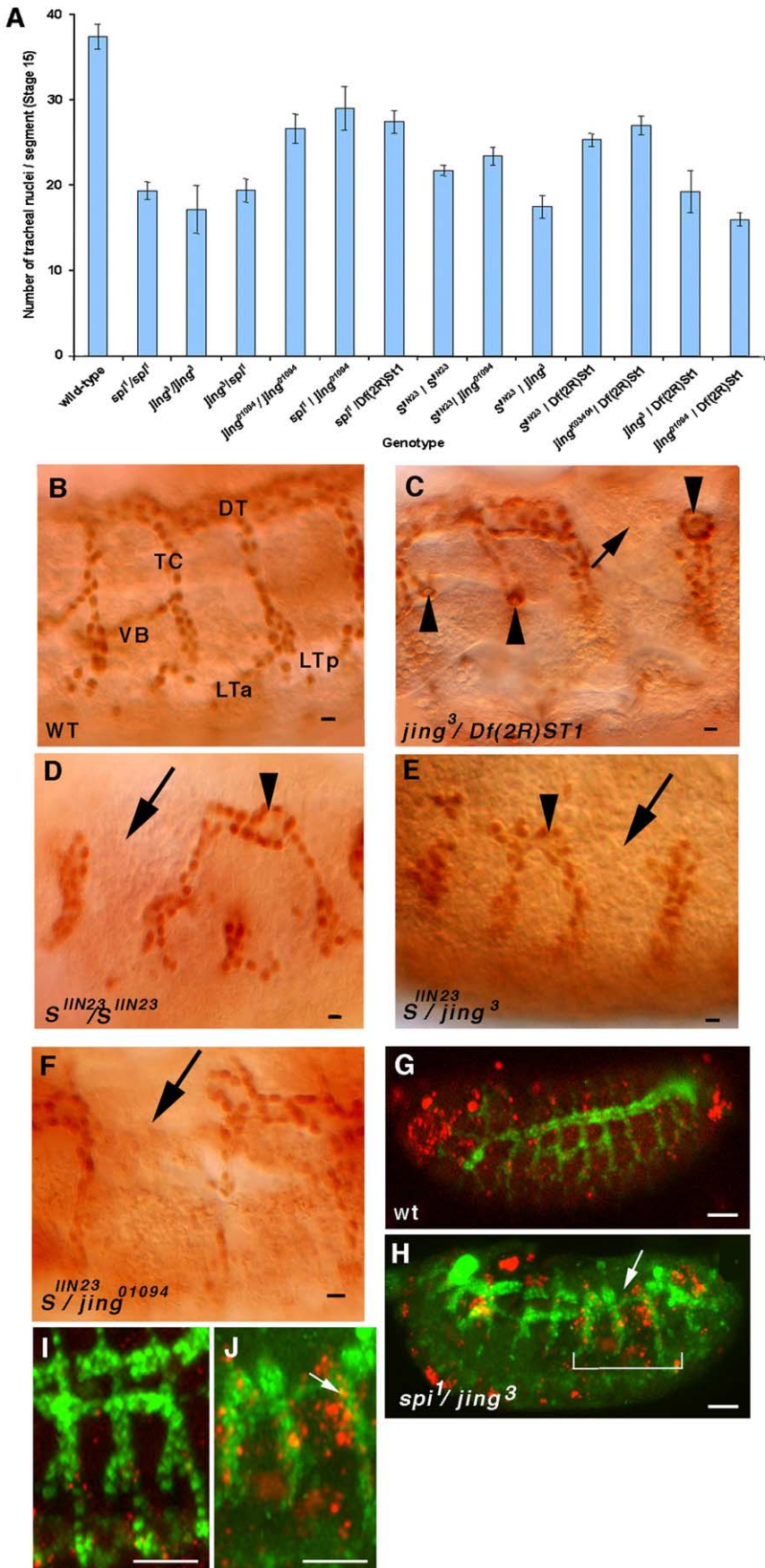
Prior work has established the important role that the *Egfr* plays during the differentiation of midline glia (MG) and tracheal cells (Klambt, 1993; Scholz et al., 1997; Sonnenfeld and Jacobs, 1994; Wappner et al., 1997; Stemerink and Jacobs, 1997). The results in this paper indicate that the *jing* gene regulates *Egfr* signaling in the MG and trachea and there are several lines of supportive evidence. First, *jing* mutant embryos fail to maintain MAPK activity and *Egfr* expression in cells that clearly have midline and tracheal identities. Second, *jing* is required for and can induce *Egfr* pathway transcription in the CNS midline and trachea. Third, *jing* over-expression promotes

midline glial survival in a similar fashion as over-expression of *Egfr* pathway genes. Fourth, *jing*-mediated over-expression phenotypes require *Egfr* pathway function in CNS midline glia and the adult eye. Fifth, a transgenic copy of either activated *ras1*, secreted *spi* or gain-of-function *Egfr* can partially rescue midline cell death in homozygous *jing* mutants. Lastly, proper dosage of both pathways is essential for survival of midline glia and for proper tracheal morphogenesis. Together, these findings suggest that *jing* functions upstream in the *Egfr/ras1* pathway. Future studies will be aimed at elucidating the nature of the relationship between *jing* and *Egfr* pathway genes and may help in the design of therapeutics to regulate over-active RTK pathways in oncogenic cells.

To our knowledge *jing* is the only gene, other than those already characterized in the *Egfr* pathway, that can promote midline glial survival. *jing* over-expression, as driven by the *sim* and *sli* promoters, induces extra midline glia that express *Egfr*, *slit* and *sim* and appear to be rescued from apoptotic fates. The extra glia are observed during stage 13 which is consistent with the timing of *Egfr* pathway-induced extra glia (Sonnenfeld and Jacobs, 1995). The absence of apoptotic glia and the wild-type midline neuronal numbers after *jing* over-expression suggests that the supernumerary glia are not likely recruited from neuronal populations and may represent glia rescued from death due to inappropriate *Egfr* expression. This effect phenocopies gain-of-function phenotypes in EGFR signaling in the CNS midline and suggests that *jing*-mediated cell survival may be carried out by the EGFR/RAS1 signaling pathway (Klaes et al., 1994; Scholz et al., 1997; Stemerink and Jacobs, 1997). In support, *jing* over-expression phenotypes in the CNS midline and eye are suppressed by reductions in *Egfr* function.

Our results suggest that *jing* is involved in both the differentiation and survival of cells in the embryonic CNS midline and trachea. In wild-type embryos, early MAPK activity controls midline glial (MG) differentiation through activation of the downstream Ets-type transcription factor *pointed* (*pnt*) (Klambt, 1993; Scholz et al., 1997). The reductions in early MAPK activity and *Egfr* expression in the midline of *jing* mutants, therefore, reveals the requirement for *jing* function in MG differentiation. The similarities in gain- and loss-of-function midline glial phenotypes between *pnt* and *jing* are consistent with this model. In *jing* mutants, reduced MAPK activity occurs in midline and tracheal cells that express the *sim* and *trh* genes, respectively, indicating that the reductions in MAPK activity are not due to a general failure in cellular differentiation (Isaac and Andrew, 1996; Wilk et al., 1996; Nambu et al., 1991).

It is possible that improper MG differentiation in *jing* mutants could be due to cells being committed to death. However, we have detected a loss of MAPK activity prior to apoptosis in the CNS midline of homozygous *jing* mutants suggesting that early MAPK inactivity in the CNS midline is independent of the apoptotic machinery. In support,



the MG initially form in MAPK mutants and it is not until later stages, which are dependent on repression of *hid*, that the MG die (Bergmann et al., 2002). MG death in *jing* mutants may be due to a combined lack of the axon-glia contacts that are necessary for MAPK-mediated inactivation of *hid* as well as from reduced MAPK activity within the MG (Bergmann et al., 2002).

3.2. Role of *jing* in the trachea

During stage 10, EGFR signaling is activated in the central region of the tracheal placode by transcription of *rhomboid* resulting in the formation of anteroposterior branches including the dorsal trunk and visceral branch (Raz and Shilo, 1992; Zak et al., 1990; Gabay et al., 1997a,b; Wappner et al., 1997; Zelzer and Shilo, 2000). Wingless (WG) signaling originates in ectodermal cells adjacent to the tracheal placodes and causes *Egfr*-induced cells to form the dorsal trunk (Chihara and Hayashi, 2000). *jing* is expressed in most tracheal cells and its protein product is localized to their nuclei suggesting that this C₂H₂-type zinc finger may have a regulatory role directly within these cells (Sedaghat et al., 2002). Additional evidence that *jing* may have a role directly in tracheal cells comes from its perturbation of tracheal morphogenesis and alteration of *Egfr/ras* pathway gene expression profiles when over-expressed specifically in the trachea (Sedaghat et al., 2002; this work).

jing affects branching morphogenesis and cellular survival in the tracheal system and its expression in the tracheal placodes coincides with that of *Egfr* pathway genes (Sedaghat et al., 2002; Bier et al., 1990; Gabay et al., 1997a,b). *jing* and *Egfr* pathway mutants have similar tracheal phenotypes which include breaks in the dorsal trunk and reduced visceral branch formation. The reductions in *Egfr*-induced cells may explain the defects in dorsal trunk formation in *jing* homozygous mutant embryos and possibly in *jing* and *Egfr* pathway double heterozygotes. Alternatively, the dorsal trunk defects may arise from perturbations in WG signaling in the ectoderm of *jing* mutants. *spitz* group tracheal mutant phenotypes do not reflect ectodermal patterning defects but this remains to be analyzed in more detail in *jing* mutants (Wappner et al., 1997).

Our results indicate that proper *Egfr* pathway and *jing* function is required for midline and tracheal cell survival. This is the first evidence, to our knowledge, of such a survival role in the trachea and requires further investigation. However, this does not rule out the possibility that other processes involved in tracheal morphogenesis are not

affected in double heterozygotes and *jing* homozygotes. Furthermore, in *jing* homozygotes and hemizygotes, truncated tubules are present in the transverse connectives suggesting that the requirement for *jing* function is more global than that of *Egfr/ras1* (Wappner et al., 1997). In support, *jing* is expressed in embryonic tissues that are not active in MAPK suggesting that *jing* has additional functions (Sedaghat et al., 2002).

Compared to other midline and tracheal-expressed genes, those of the *Egfr* pathway are more highly expressed after *jing* over-expression but not more than three-fold. Nevertheless, the effects of *jing* over-expression in the CNS midline can be seen by extra glia and *Egfr* expression establishing the importance of regulating *jing* expression during embryogenesis. Ectopic expression analyses suggest that *jing* is not sufficient to activate *Egfr* pathway gene expression. Therefore, these results suggest that in order to induce gene expression *jing* may require another protein, such as a cell-specific chromatin remodeling protein, that is not present in *prd* stripes but is present in the CNS midline, trachea and eye. The exact relationship between *jing* and *Egfr* pathway genes requires further analysis.

4. Experimental procedures

4.1. *Drosophila* stocks

spitz (*spi*¹) (Nüsslein-Volhard et al., 1984), *Star* (*S*^{IN23}) and the *jing* deficiency, *Df*(2R)*ST1* (Liu and Montell, 2001; Sedaghat et al., 2002), were obtained from the Bloomington *Drosophila* stock center. *Df*(2R)*Egfr5* (Price et al., 1989) and *Df*(2R)TW50 (Butler et al., 2001) are deficient in *Egfr* and *spi*, respectively, and were obtained from the DK2 kit (Bloomington *Drosophila* stock center). *jing*³, *jing*⁰¹⁰⁹⁴ and *jing*^{K03404} were previously described (Sedaghat et al., 2002). *w*¹¹⁸ was used as a control strain. The *slit* (*sli*) reporter P[*sli* 1.0 HV-*lacZ*] was used to assess transcription of the *sli* gene (Rothberg et al., 1988; Wharton et al., 1994; Ma et al., 2000).

The following *Gal4* driver lines were used: P[*sim-Gal4*] (second chromosome; provided by S. Crews); P[*sli-Gal4*] (second chromosome; provided by R. Jacobs), P[*prd-Gal4*] and P[*GMR-Gal4*] (provided by J. Nambu; Ma et al., 2000); P[*btl-Gal4*] (Shiga et al., 1996). The following *UAS* lines were used: P[*UAS-lacZ*]; P[*UAS-sspi4a*] (Golembo et al., 1996b); P[*UAS-jing*] (Sedaghat et al., 2002); P[*UAS-ras*^{Val12}]

Fig. 8. The *jing* and *Egfr* pathways display dominant genetic interactions in the trachea. Sagittal views of stage 15 embryos stained with anti-TRH are shown. (A) Quantitative analysis of tracheal nuclei in *jing* hemizygotes and *jing* and *Egfr* pathway double heterozygotes. (B) The wild-type trachea. (C–F) There are breaks in the dorsal trunk (arrows) in embryos hemizygous for *jing* (C), homozygous for *S*^{IN23} (D) and double heterozygous for *S*^{IN23} and *jing*³ (E) or *jing*⁰¹⁰⁹⁴ (F). Arrowheads denote unfused dorsal trunk and truncated transverse connectives. (G–J) Stage 15 embryos were stained with anti-TRH and TUNEL and viewed by confocal microscopy. (G) Wild-type whole-mount embryo. (H) A *spi* and *jing* double heterozygote showing breaks in the dorsal trunk (arrow). (I) Close-up view of wild-type trachea. (J) Close-up view of area denoted in (H). Note presence of TRH + TUNEL + cells (arrow). Scale bars in (B)–(F) are 10 μm and in (G)–(I) are 30 μm. DT, dorsal trunk; TC, transverse connective; VB, visceral branch; LTa, lateral trunk anterior; LTP, lateral trunk posterior. Fluorescent images are merged.

(Fortini et al., 1992); and P[UAS-ellipse] (provided by J.R. Jacobs).

4.2. Genetics and generation of rescue lines

All crosses were performed at 25 °C. In rescue experiments, *lacZ* marked balancer chromosomes were used to distinguish transgenic embryos lacking endogenous *jing* from those that carried a functional copy of the *jing* gene (supplied by the balancer chromosome). For rescue in homozygous *jing* mutant backgrounds two stable lines were produced. In the first line, P[*sim-Gal4*] was recombined onto *jing* mutant chromosomes by meiotic recombination and balanced with *CyO wg-lacZ*. To generate the second stock, P[UAS-*jing*] was incorporated into *jing* mutant backgrounds by standard genetic techniques. Single-copy rescue was assessed by collecting embryos from a cross of P[UAS-*jing*]; *jing*/CyO *wg-lacZ* females to P[*sim-Gal4*] *jing*/CyO *wg-lacZ* males.

jing was over-expressed in the CNS midline using the *sim-Gal4* and *sli-Gal4* drivers and two copies of P[UAS-*jingE*] or P[UAS-*jingU*]. For targeted expression of *Egfr* pathway genes in homozygous *jing* mutant embryos, P[*sim-Gal4*] *jing*/CyO *wg-lacZ* females were crossed to UAS-*ras*^{V12} *jing*/CyO *wg-lacZ*, P[UAS-ellipse] *jing*/CyO *wg-lacZ* or P[UAS-*sspi4a*] *jing*/CyO *wg-lacZ* males. For over-expression of *jing* in *S* and *spi* homozygous mutant backgrounds, the P[*sli-Gal4*] driver was recombined onto the *S* and *spi* mutant chromosomes and P[UAS-*jing*] was incorporated into these fly strains by standard genetic techniques. In addition, embryos of the genotype P[UAS-*jing*]; *spi*/wg-*lacZ* CyO were generated. Embryos containing P[*sli-Gal4*], two copies of P[UAS-*jing*] and that were homozygous for *S* or *spi* were identified by their lack of expression of the *CyO wg-lacZ* balancer.

4.3. RNA isolation, microarray analysis and in situ hybridization

A microarray screen was performed to identify *jing* transcriptional targets (in preparation). Total RNA was prepared from control and experimental embryonic samples using an RNeasy kit (Qiagen). All embryos were collected at 25 °C. The RNA samples were labeled independently with Cy3 and Cy5 and then hybridized to an array by the Canadian *Drosophila* Microarray Centre (CDMC). The array includes the Berkeley DGC EST set (approximately 5900 clones), B. Oliver's and J. Andrews non-DGC clones (approximately 750 clones from a testis library), genes from submitted lists (approximately 572 genes).

For control over-expression experiments, flies carrying *btl-Gal4* and one copy of UAS-*lacZ* were crossed to UAS-*lacZ* and RNA was extracted from their embryonic progeny. To analyze *jing* over-expression, RNA was extracted from embryos of a cross between flies carrying *btl-Gal4* and one copy of UAS-*jing* and flies carrying UAS-*jing*. One-half

of all progeny over-expressed *jing* in the CNS midline and trachea whereas 1/4th expressed two copies. Four independent samples from each group were analyzed. In all samples, embryos were aged to 7–9 h after egg laying at 25 °C.

For loss-of-function experiments, the control samples included RNA extracted from *w*¹¹⁸ embryos to ensure a similar genetic background as the *jing* mutants (Sedaghat et al., 2002). Experimental samples included RNA extracted from embryos homozygous mutant for *jing*³. Values represent averages of four independent samples. Embryos were aged as described above and un-fixed homozygous embryos were selected after staining with X-Galactosidase. Control *w*¹¹⁸ embryos were treated in a similar fashion. Microsoft excel was used to calculate the mean, standard error and statistical significance for the Cy3/Cy5 ratios of *Egfr* pathway genes.

In situ hybridization was carried out using digoxigenin-labeled riboprobes prepared from *Egfr* (Stemerdink and Jacobs, 1997) or *jing* cDNA (Sedaghat et al., 2002) according to Janody et al. (2000). Whole-mount embryos were collected from control *w*¹¹⁸, *jing*³, *jing*⁰¹⁰⁹⁴, P[*prd-Gal4*]/P[UAS-*jingE*] and P[*sli-Gal4*]; P[UAS-*jingE*]/P[UAS-*jingE*]. The number of embryos with loss of *Egfr* expression was counted in experimental and control populations.

4.4. Antibodies

Monoclonal antibody BP102 was obtained from the Developmental Studies Hybridoma Bank. The following antibodies were used; mAb anti-β-galactosidase (anti-β-gal) (Promega); rabbit polyclonal anti-β-gal (Promega); rat polyclonals anti-Single-minded and anti-Trachealess (obtained from S. Crews) (Sonnenfeld et al., 1997; Ward et al., 1998). Anti-diphospho-MAPK was obtained from Sigma (Gabay et al., 1997a,b).

4.5. Immunohistochemistry, confocal microscopy and TUNEL labeling

Embryo staging was according to Campos-Ortega and Hartenstein (1997) as modified by Klämbt and Goodman (1991). Processing for light microscopy was according to standard protocols (Patel, 1994). Balancer chromosomes carrying *lacZ* for the second (*CyOP[wg-lacZ]*) and third chromosomes (*TM3P[ubx-lacZ]*) were used to identify homozygous mutant embryos after anti-β-Gal staining. Antibody staining was visualized using HRP-, FITC- or rhodamine-conjugated secondary antibodies (Jackson). HRP-labeled embryos were analyzed by light microscopy using a Zeiss Axioskop. In the case of anti-diphospho-MAPK (Gabay et al., 1997a,b), peroxidase staining was enhanced using Tyramide Signal Amplification reagents (TSA, New England Nuclear). Fluorescent labeling was visualized on a Zeiss Axiovert 100 TV confocal microscope.

TUNEL (TMR red; Roche) staining was carried out according to previous procedures and was double stained with anti-Sim, anti-TRH or anti- β -GAL (Booth et al., 2000). Fluorescently labeled embryos were mounted in 4% *n*-propyl gallate to inhibit photobleaching and analyzed on a Zeiss Axiovert 100 TV confocal microscope. Optical sections of 1 μ m were recorded in line average mode and parameters were set by the TRH and SIM fluorescence. All figures were processed with Adobe Photoshop software.

4.6. Genetic interactions in the eye

To analyze genetic interactions in the eye, females carrying P[GMR-Gal4] and P[UAS-*jing*] were crossed to males carrying; (1) *spi*¹; P[UAS-*jing*], (2) Df(2R)*Egfr*5; P[UAS-*jing*] or (3) Df(2L)TW50; P[UAS-*jing*]. In addition, females carrying P[GMR-Gal4] and P[UAS-*jing*] were crossed to males carrying P[UAS-*ellipse*] and compared to progeny from crosses of P[GMR-Gal4] and P[UAS-*ellipse*] or P[UAS-*jing*]. Rough eye phenotypes were identified under a dissecting microscope at 25 °C where 1/16th of progeny carried two copies of P[UAS-*jing*] and one copy of P[GMR-Gal4] and *spi* or Df(2R)*Egfr*5 or Df(2L)TW50. Crosses were repeated in triplicate.

The ommatidial arrangement was analyzed in images generated by scanning electron microscopy and light microscopy by the bioimaging centre at Mount Sinai Hospital (Toronto, Canada). For light and scanning electron microscopy, eye specimens were fixed in 2% glutaraldehyde in 0.1 M phosphate buffer, washed in buffer, then transferred into liquid 4% agarose. After the agarose cooled and solidified the eyes were cut out in cubes and dehydrated in a graded ethanol series followed by propylene dioxide and embedded in Spurr epoxy resin in flat molds. One micrometer serial sections were cut with glass knives on a RMC MT6000 ultramicrotome and were stained with Toluidine Blue. The number of ommatidia in light-level sections was determined in two 30 μ m areas in each section. Three sections from independent eyes were scored.

4.7. Quantification of CNS midline and tracheal cells

To quantify midline cells, the MG were identified using the P[*sli* 1.0 HV-*lacZ*] reporter whereas both MG and neurons were identified using anti-Sim antibody. For over-expression and rescue experiments, the number of MG per nerve cord segment was determined by light microscopy of HRP-stained stage 13 embryos. Midline cells labeled with MAPK and anti-SIM were counted on confocal microscopic images. To determine the effects of *ras*1 and *Egfr* rescue on midline cell survival, the number of midline cells in homozygous and transheterozygous *jing* mutants was determined using *sim-GAL4/UAS-lacZ* as a marker of cell identity. Microsoft Excel and unpaired *t* tests were used to determine the statistical significance compared with *w*¹¹⁸ values.

To determine the effects of *jing* and the *Egfr* pathway on cell survival, the number of midline cells in homozygous and double heterozygous embryos was counted in stage 15 embryos stained with anti-SIM. At least 30 segments were scored in each case. The number of TRH+ cells was determined in each tracheal segment in stage 14/15 *jing* homozygotes, transheterozygotes and *jing* and *spi* or *S* double heterozygotes. At least 40 segments were scored in each case. Microsoft Excel and unpaired *t* tests were used to determine the statistical significance compared with *w*¹¹⁸ values.

To determine the extent of apoptosis in the CNS midline, the number of SIM⁺ and TUNEL⁺ cells (yellow cells) was counted per embryo for a particular genotype. To analyze apoptosis in the trachea, the number of TRH⁺ and TUNEL⁺ cells was counted per tracheal segment in stage 14/15 embryos. In each case, the average, standard error and statistical significance were calculated using Microsoft Excel for at least 50 tracheal segments and VNC segments. Scanning parameters during confocal microscopy were set by the SIM and TRH fluorescence.

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